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CoordiCEPS: A Coordinated Cross-Source Indirect Prompt Injection Benchmark for LLM Agents

PDF (/pdf?id=XmSXxacdb3)

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Keywords: Indirect Prompt Injection, LLM Agents, Agent Security, Adaptive Injection Attack, Cross-Source Attack
TL;DR: CoordiCEPS is an executable benchmark for evaluating tool-using agents against coordinated indirect prompt injections that distribute malicious intent across multiple data sources in an environment.
Abstract:
The proliferation of tool-using large language model (LLM) agents increases exposure to indirect prompt injection, in which malicious instructions embedded in external data sources manipulate agent behavior. Existing benchmarks have progressed from single-source injections to multi-site and long-horizon attacks, but do not systematically evaluate how multiple injected sources coordinate to steer an agent toward a common malicious objective. We present CoordiCEPS, a benchmark for coordinated cross-source indirect prompt injection in tool-using agents. CoordiCEPS defines an attack strategy space along two complementary dimensions: impetus, which specifies why an agent should follow the injection task, and coordination, which specifies how multiple sources jointly advance that task. The benchmark contains 2,400 executable attack instances spanning four application scenarios, six attack strategies, and the Cartesian product of 10 user tasks and 10 injection tasks per scenario. Each instance is constructed through a two-stage procedure that first develops a plan to induce deviation from the benign trajectory by integrating selected carriers into a strategic narrative, and then drafts injection payloads, or snippets, according to their assigned roles. Experiments on six LLMs and four defenses reveal substantial variation across scenarios, attack strategies, models, and defense mechanisms. Even the most robust model remains vulnerable, and no evaluated defense consistently achieves both low attack success and high secure utility across all scenarios. These results demonstrate that coordinated cross-source injections remain a nontrivial challenge for current tool-using agents.

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Number Of Submissions: I agree that none of the authors have more than two submissions in this cycle of the KDD D&B track.
Submission Number: 279

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Official Review of Submission279 by Reviewer 9XVo

Official Review by Reviewer 9XVo 10 Sept 2026, 17:56 (modified: 27 Sept 2026, 22:52)
Program Chairs, Area Chairs, Reviewers Submitted, Authors Revisions (/revisions?id=HpImVLI4Bm)

Paper Summary:
This paper introduces CoordiCEPS, a benchmark for coordinated cross-source indirect prompt injection attacks on tool-using LLM agents. The benchmark defines six attack strategies using two dimensions: impetus (dependency or priority) and coordination (repetition, handoff, or breakdown). Attacks are generated with a trajectory-grounded plan-then-write pipeline that selects multiple reachable carriers and writes coordinated injection content for them. The final benchmark contains 2,400 instances across Slack, Workspace, Travel, and Banking, covering the Cartesian product of 10 user tasks, 10 injection tasks, and six attack strategies per scenario. The benchmark is evaluated on six LLMs and four lightweight defenses. The results show large differences across models, scenarios, and coordination strategies, suggesting that the benchmark is useful for distinguishing agent robustness.

Paper Strengths:
1. Good benchmark design and clear track fit. The paper isolates a concrete gap in existing IPI evaluation: multi-source attacks have been studied, but the coordination structure itself is rarely varied systematically. The impetus × coordination taxonomy is simple and useful.
2. Systematic coverage. Each scenario contains all 100 user-task/injection-task pairs under all six strategies, giving 600 instances per scenario and 2,400 overall. This is cleaner than selecting a small set of hand-designed attacks.
3. The benchmark produces meaningful model separation. Vulnerability varies substantially across models and scenarios. For example, GPT-5.5 is relatively robust, while DeepSeek-V4-Pro is much more vulnerable. The relative strength of coordination strategies also changes by scenario rather than collapsing to one universally dominant attack.
4. The benchmark is executable rather than prompt-only. Success is evaluated through changes to simulated environment state, which is substantially more informative than judging attack text in isolation. The appendix also gives a useful end-to-end example showing the attack plan, generated carriers, and resulting victim trajectory.

Paper Weaknesses:
1. The paper does not isolate the benefit of coordinated cross-source attacks. The central motivation is that coordinated attacks expose vulnerabilities not captured by existing single-source or fixed-structure attacks. However, the experiments only compare the six CoordiCEPS strategies against each other. There is no matched single-source, uncoordinated multi-source, or AgentLAB-style baseline on the same task pairs.

2. Generation is tuned to one victim model. All instances are generated using DeepSeek-V4-Pro as the attacker and GLM-5.2 as the victim. Failed attacks are regenerated up to three times, and later attempts may use previously successful attacks as in-context examples.
3. The evaluation does not fully establish the realism of the generated attacks. The generator receives the ground-truth user trajectory, tool outputs, carrier metadata, and victim-model information. This is a fairly strong attacker-information assumption. It is reasonable for constructing a stress-test benchmark, but the threat model and its implications for real-world representativeness deserve more discussion.
4. Strategy comparisons lack uncertainty estimates. Each strategy/scenario cell contains 100 task pairs, but Tables 1–5 report point estimates only. Some differences between strategies are small, yet the paper interprets them as differences in coordination effectiveness. Since the same user/injection task pairs are evaluated across strategies, paired bootstrap confidence intervals or paired tests would be straightforward and would make these comparisons more convincing.

Benchmark Quality: 3: Good Benchmark - useful benchmark with some challenges and generalizability, likely to be adopted but with limitations in scope, coverage, or long-term impact.
Relevance: 4: High - The dataset or the benchmark address an important problem in machine learning and is of broad interest to the community
Novelty: 3: Moderate - The dataset is moderately novel and introduces new and interesting data types, tasks, or problem formulations. The contribution is original and represents an advancement of existing knowledge.
Technical Quality: 4: High - The dataset is well-curated, clean, and representative of real-world scenarios. The data collection and evaluation procedures are well-documented and reproducible. Any biases or limitations are appropriately discussed.
Usability And Accessibility: 3: Moderate - The dataset and benchmarks are accessible with some effort. The code, scripts, and metadata can be understood with some effort.
Reproducibility: 3: Moderate - The paper provides a clear and detailed description of the methods, data, and procedures used. Supplementary materials, such as datasets and code, are available and sufficiently documented. Reproducing the results would be feasible with the provided information, though some effort may still be required.
Review Rating: +4: Accept
Reviewer Confidence: 3: Moderate - The reviewer has a good understanding of the subject area and is familiar with the research methods and context of the paper. They feel confident in their ability to accurately assess the quality and significance of the work. Their evaluation is based on a solid grasp of the content and context.
Ethics Review Flag: No
Ethics Review Description:
NA

Official Review of Submission279 by Reviewer Dm8i

Official Review by Reviewer Dm8i 09 Sept 2026, 19:03 (modified: 27 Sept 2026, 22:52)
Program Chairs, Area Chairs, Reviewers Submitted, Authors Revisions (/revisions?id=V3Xt8LyEJN)

Paper Summary:
In this paper the authors propose CoordiCEPS, a benchmark for coordinated cross source indirect prompt injection in tool calling agents. CoordiCEPS defines an attack strategy space along two complementary dimensions, impetus, which specifies why an agent should follow the injection task, and coordination, which specifies how multiple sources jointly advance that task.

Paper Strengths:
- The paper proposes a benchmark generation process and an benchmark dataset that enables the evaluation of the robustness of tool calling agents against coordinated cross source indirect prompt injection attacks. Which makes evaluation of agents robustness against more complex, realistic style of prompt injection attack.
- This work demonstrates from empirical results that tool calling agents need more internal (at the model level) and external defenses against cross source injection attacks.

Paper Weaknesses:

Weaknesses:

- The authors cite that two of the major contribution of this work is the injection generation pipeline but the technical discussions in a number of sections on the generation process are lacking much needed details. This lack of details weakness the paper and makes it a bit harder to see the novelty of the contributions. Please see suggestions and questions below for more details.

Suggestions:

- Section 1: While you do cite a source with a general statement about the increase in malicious prompt injections increasing, if possible including some concrete metrics would make the case stronger that this kind of benchmark serves an immediate need.
- Section 3: Having to jump to section 4.3 from Section 3 to understand what variable z is, creates a gap in understanding the problem definition fully since the discussion for a key variable is split across different sections in the paper. Even if you do not include the full discussion, a bit more of information on the purpose the varible serves would give the reader more context, which would help strengthen the discussion.
- Section 4.3: This is a very dense section, the use of diagrams and/or more structured formated text like bullet point lists could improve the readability of this section.
- Section 4.3: In this section you mention the attacker is using an LLM, a more formal discussion and walkthrough of how the LLM is used from an attackers perspective would be helpful here. The way this section is currently written it seems to make an assumption regarding how familiar the reader is with this topic.
- Figure 1: While this figure is high level diagram of the injection generation pipeline, it does match the walkthrough of the benchmark generation process a described in Section 4. If the subsections in Section 4 and the parts of the diagram were more aligned, this would make the discussion regarding how the benchmark is generated a bit more clear.

Questions:

- Abstract/Section 1: How common is this relatively complex injection attack scenario currently in public web? Adding metrics that show an uptick, trend of prevalence would strengthen the discussion of the motivation for your research.
- Section 4.4: Does this rerun go indefinitely or is there a configured or hard coded limit? Adding more details about this process would make the discussion clearer.

Benchmark Quality: 3: Good Benchmark - useful benchmark with some challenges and generalizability, likely to be adopted but with limitations in scope, coverage, or long-term impact.
Relevance: 4: High - The dataset or the benchmark address an important problem in machine learning and is of broad interest to the community
Novelty: 3: Moderate - The dataset is moderately novel and introduces new and interesting data types, tasks, or problem formulations. The contribution is original and represents an advancement of existing knowledge.
Technical Quality: 3: Moderate - The dataset demonstrates solid technical quality with a sound methodology and thorough analysis. The data collection and evaluation procedures are reliable and well-supported. There may be minor issues, but they do not significantly undermine the overall quality. The work is competently executed and meets acceptable standards.
Usability And Accessibility: 3: Moderate - The dataset and benchmarks are accessible with some effort. The code, scripts, and metadata can be understood with some effort.
Reproducibility: 3: Moderate - The paper provides a clear and detailed description of the methods, data, and procedures used. Supplementary materials, such as datasets and code, are available and sufficiently documented. Reproducing the results would be feasible with the provided information, though some effort may still be required.
Review Rating: -2: Weak Reject
Reviewer Confidence: 3: Moderate - The reviewer has a good understanding of the subject area and is familiar with the research methods and context of the paper. They feel confident in their ability to accurately assess the quality and significance of the work. Their evaluation is based on a solid grasp of the content and context.
Ethics Review Flag: No
Ethics Review Description:
N/A

Official Review of Submission279 by Reviewer xyRf

Official Review by Reviewer xyRf 08 Sept 2026, 22:56 (modified: 27 Sept 2026, 22:52)

Program Chairs, Area Chairs, Reviewers Submitted, Authors Revisions (/revisions?id=JTn3k46HvR)

Paper Summary:
This paper presents CoordiCEPS, a benchmark for studying coordinated cross-source indirect prompt injection attacks against tool-using LLM agents. The authors argue that existing benchmarks often evaluate attacks from individual malicious sources, while realistic attackers may distribute instructions across multiple sources to gradually influence agent behavior. To address this gap, the paper proposes a taxonomy based on attack motivation and coordination patterns, and constructs executable attack cases across multiple agent environments. The authors evaluate different LLM agents and defense mechanisms, showing that coordinated injections remain challenging for current systems.

Paper Strengths:
1. The paper identifies a meaningful limitation of existing indirect prompt injection evaluations. Studying how multiple external sources cooperate toward a shared malicious goal provides a more challenging setting than conventional single-source injection attacks.
2. The proposed attack categorization is useful for organizing different coordination behaviors. It allows researchers to analyze whether certain forms of source cooperation are more effective than others.
3. The benchmark provides relatively broad coverage, including multiple application scenarios, attack strategies, models, and defense mechanisms. This makes the evaluation more comprehensive than manually designed attack examples.
4. The attack generation procedure is a practical contribution. The two-stage design attempts to preserve consistency between different injected sources while creating realistic agent trajectories.

Paper Weaknesses:
1. The main concern is whether automatically generated attacks accurately represent real-world prompt injection threats. Although the generated examples are designed to be realistic, stronger validation against naturally occurring attacks would improve confidence in the benchmark.
2. The benchmark assumes that attackers can coordinate multiple sources and construct consistent injection strategies. However, the practical cost and required knowledge for such attacks are not deeply analyzed.
3. The evaluation mainly reports overall attack success and defense performance. More detailed analysis of why certain coordination strategies bypass specific defenses would make the conclusions more informative.
4. The current benchmark is evaluated within predefined simulated environments. Additional experiments on unseen scenarios or real-world agent applications would help demonstrate the general applicability of the findings.
5. Some conclusions may depend on the capability of the attacker-side generation model. A stronger discussion about attacker model selection and its influence on benchmark difficulty would strengthen the paper.

Benchmark Quality: 4: Strong Benchmark - A difficult-to-create benchmark (dataset, framework, or both) addressing an important problem, enabling broad algorithm evaluation, and having high potential for widespread use.
Relevance: 4: High - The dataset or the benchmark address an important problem in machine learning and is of broad interest to the community
Novelty: 3: Moderate - The dataset is moderately novel and introduces new and interesting data types, tasks, or problem formulations. The contribution is original and represents an advancement of existing knowledge.
Technical Quality: 3: Moderate - The dataset demonstrates solid technical quality with a sound methodology and thorough analysis. The data collection and evaluation procedures are reliable and well-supported. There may be minor issues, but they do not significantly undermine the overall quality. The work is competently executed and meets acceptable standards.
Usability And Accessibility: 4: High - The dataset and benchmarking are easily accessible and well-documented. The code, scripts, and metadata are provided for easy use. The dataset complies with legal and privacy considerations.
Reproducibility: 4: High - The data collection procedures are well-documented and reproducible. Any biases or limitations are appropriately discussed. The evaluation setup is robust and aligned with best practices. Supplementary materials, including datasets and code, are complete, well-documented, and easily accessible. Reproducing the results would be straightforward and require minimal additional effort.
Review Rating: +4: Accept

Reviewer Confidence: 3: Moderate - The reviewer has a good understanding of the subject area and is familiar with the research methods and context of the paper. They feel confident in their ability to accurately assess the quality and significance of the work. Their evaluation is based on a solid grasp of the content and context.
Ethics Review Flag: No
Ethics Review Description:
No ethics review flag is raised.

Official Review of Submission279 by Reviewer WQ1L

Official Review by Reviewer WQ1L 20 Aug 2026, 00:57 (modified: 27 Sept 2026, 22:52)
Program Chairs, Area Chairs, Reviewers Submitted, Authors Revisions (/revisions?id=BH49aE8lJv)

Paper Summary:
This paper introduces CoordiCEPS, a benchmark for evaluating the robustness of tool-using LLM agents against coordinated indirect prompt injection attacks distributed across multiple external sources. The key contribution is a systematic attack strategy matrix that varies attacks along two dimensions—impetus (why the agent follows the injection) and coordination (how multiple sources jointly advance the attack)—combined with a two-stage generation procedure that produces realistic, executable attack instances across four application scenarios. The benchmark includes 2,400 instances and evaluation of six LLM agents and four defenses, revealing that even the most robust models remain vulnerable, with attack success rates ranging from 19.2% to 89.2%, and that no evaluated defense consistently achieves both low attack success and high secure utility. The work is significant because it exposes a critical blind spot in existing IPI benchmarks—coordinated multi-source attacks—and provides a standardized framework for systematically evaluating and improving agent robustness against this emerging threat.

Paper Strengths:
1. The paper's systematic decomposition of multi-source coordination into two simple, orthogonal dimensions—impetus and coordination—provides a clean, generalizable framework that can inspire future benchmarks to move beyond monolithic attacks and explore structured adversarial narratives across diverse agent environments.
2. The stark empirical finding that even the most robust frontier model remains vulnerable (ASR 19.2%) while all evaluated defenses show inconsistent tradeoffs between attack resistance and utility underscores a critical, sobering result for the community: coordinated cross-source injections pose a fundamentally harder problem that current isolated-payload benchmarks and lightweight defenses fail to adequately address.

Paper Weaknesses:
1. The paper positions CoordiCEPS as a substantial advance over existing benchmarks, yet its entire technical foundation—executable scenarios, tool interfaces, task sets, evaluation infrastructure, and even the regeneration method using successful instances as in-context examples—is explicitly built upon and adapted from AgentLAB. The proposed “coordination” taxonomy largely re-describes the multi-observation, long-horizon attack structures that AgentLAB’s Task Injection already implements, but without a clear empirical or formal demonstration that these particular coordination patterns expose fundamentally new failure modes beyond what AgentLAB already covers. While the systematic Cartesian product over strategies is a contribution, the core generation pipeline is described as "based on a method similar to that in AgentLAB," which significantly diminishes the perceived novelty and makes the contribution feel more like a moderate extension than a breakthrough.
2. The attack generation procedure crucially relies on the ground-truth user-task trajectory to identify which carriers are reachable and where injections should be placed. This gives the attacker omniscient knowledge of the agent’s exact future tool calls, observations, and execution path. In real-world deployments, an attacker does not have such perfect foresight; they must guess which sources a diverse set of users might interact with and in what order. While the authors frame this as a “worst-case” evaluation, the paper does not sufficiently discuss this limitation or provide a sensitivity analysis (e.g., what happens if the planner is given only partial trajectory information, or if the agent deviates from the expected path). Consequently, the reported attack success rates likely overestimate real-world vulnerability, and the benchmark’s representativeness of practical threat scenarios is questionable.

3. The paper concludes that defenses have inconsistent utility tradeoffs, but it only evaluates four lightweight and arguably outdated baselines: PromptArmor, repeated prompting, spotlighting, and tool filtering. The authors explicitly acknowledge in Section 2.2 that multiple stronger defenses exist—Firewall, Task Shield, DRIFT, MELON, CausalArmor, ARGUS, and CaMeL—which impose interface filtering, causal attribution, trajectory validation, or architectural separation. However, these are deliberately excluded from evaluation with the hand-wavy justification that they are "complementary" or require "deeper integration." This is a major gap: concluding that *lightweight* defenses are insufficient does not demonstrate that coordinated attacks pose a fundamental, unsolved challenge. To convincingly show the benchmark's severity, the authors must test at least one or two of these stronger, more representative defenses and show that they also fail. As it stands, the defensive conclusions are both expected and uninformative.
4. While the paper extensively reports that ASR varies across coordination strategies—e.g., Priority+Handoff often achieves high ASR while Dependency+Breakdown is sometimes lower—it provides virtually no *mechanistic* analysis of *why* these differences occur. Does Handoff succeed because of increased narrative coherence, because it forces the agent to take more steps before detecting the attack, or because it bypasses certain state checks? Does Breakdown fail in some scenarios because agents can recover the full objective from partial information, or because required carriers are not all read? The paper offers no ablation studies on carrier count, ordering, or semantic distance, and no qualitative analysis of failure cases across strategies. This reduces the benchmark from an analytical tool that could inform future defenses into a mere data-reporting exercise, limiting its impact on the research community.

Benchmark Quality: 2: Limited Benchmark Value - Includes a benchmarking component but is narrow in scope, lacks generality, or has limited potential for widespread adoption.
Relevance: 3: Moderate - The work is somewhat relevant to KDD and is of narrow interest to a sub-community
Novelty: 2: Low - The contributions are relatively minor and largely incremental. The work builds heavily on existing datasets and benchmarks.
Technical Quality: 2: Low - The dataset, data collection, and evaluation procedures have several technical weaknesses, such as methodological flaws, insufficient analysis, or unsupported conclusions. While the work shows some level of competence, it lacks thoroughness and precision. Improvements are necessary for it to be considered robust.
Usability And Accessibility: 3: Moderate - The dataset and benchmarks are accessible with some effort. The code, scripts, and metadata can be understood with some effort.
Reproducibility: 3: Moderate - The paper provides a clear and detailed description of the methods, data, and procedures used. Supplementary materials, such as datasets and code, are available and sufficiently documented. Reproducing the results would be feasible with the provided information, though some effort may still be required.
Review Rating: -4: Reject
Reviewer Confidence: 3: Moderate - The reviewer has a good understanding of the subject area and is familiar with the research methods and context of the paper. They feel confident in their ability to accurately assess the quality and significance of the work. Their evaluation is based on a solid grasp of the content and context.
Ethics Review Flag: No
Ethics Review Description:
no

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